Chapter 26

Single-Cell Proteomics: Low-Input Quantification, scp Data Architecture, and Sparse Matrix Modeling

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Abstract

Bulk quantitative proteomics measures the ensemble average of millions of cells, masking stochastic transcriptional bursting, rare functional sub-populations, and cell-to-cell biochemical heterogeneity. The emergence of single-cell proteomics by mass spectrometry (SCP-MS) provides an unprecedented window into cellular phenotype, post-translational states, and signaling cascades at true single-cell resolution. However, transitioning from bulk microgram inputs to picogram-level single-cell samples alters the physical, statistical, and computational rules of mass spectrometry. Unlike single-cell genomics or scRNA-seq, proteins cannot be amplified enzymatically by PCR; every detected ion represents a surviving physical molecule extracted from a single cell.

*Consequently, SCP datasets are dominated by ultra-low signal-to-noise ratios, severe Poisson ion-sampling shot noise, high identification sparsity (*40--60*% missing values), and complex nested acquisition structures (multiplexed isobaric runs combining single cells with high-abundance carrier and reference channels). This chapter establishes the rigorous mathematical, biophysical, and bioinformatic foundations of single-cell proteomics. We formalize the physics of ion transmission efficiency, analyze the carrier channel paradigm and its quantitative trade-offs, and dissect the* `scp` *framework, which extends Bioconductor's* `QFeatures` *and* `SingleCellExperiment` *classes into a unified multi-assay data tree. We examine cell-level quality control, Tukey biweight feature aggregation, leakage-free normalization, and the acute hazards of naive missing-data imputation. Following the hands-on data analytics pedagogy [5], the chapter culminates in an executable, end-to-end R laboratory, the ''Five Deadly Sins'' of single-cell proteomics, a self-assessment diagnostic checklist, and multi-tier exercises with step-by-step worked solutions.*

Keywords: Single-Cell Proteomics (SCP), scp Package, QFeatures, Isobaric Carrier Proteome, Poisson Shot Noise, SingleCellExperiment, Matrix Sparsity, Pseudoreplication, timsTOF SCP, Orbitrap Astral.

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Table 26.1: Tutor's Methodological Roadmap: Four Checkpoints for Single-Cell Proteomics Analysis

Stage	Physical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Ion Trapping & Carrier	Sub-nanogram inputs (100--200 pg/cell); trap charge capacity $C_{\text{cap}} \approx 10^5$--$10^6$ charges [25, 8].	Balance carrier amplification against space-charge single-cell ion suppression; quantify Poisson limit.	Excessive carrier loading ($> 200\times$) suppresses single-cell signal into noise. Remedy: Restrict booster to 50--100$\times$ equivalents [30].
2. Hierarchical Container	Multi-level data topology: PSMs $\rightarrow$ Peptides $\rightarrow$ Proteins across multiplex runs [13, 31].	Structure multi-assay containers linking raw reporter ions to aggregated protein groups via `scp`.	Flattish data matrices discarding peptide-to-protein provenance. Remedy: Implement `QFeatures` tree architecture [32].
3. Cell QC & Normalization	Droplet dispensing variations, empty wells, cell doublets, and run-to-run batch effects [33, 14].	Audit channel signal-to-blank ratio (> 3.0), filter doublets, and execute leakage-free median/IRS scaling.	Normalizing single cells while retaining carrier channel. Remedy: Subsetting single cells via `subsetByColData()` before scaling [22].
4. Sparsity & Inference	Hybrid MAR/MNAR missingness (40--60%) and cell-level pseudoreplication [2, 3, 6].	Perform dimensionality reduction on complete cases without bulk imputation; model donor clustering via mixed models.	Imputing high-sparsity matrices with KNN, manufacturing false clusters. Remedy: Non-imputed filtering and mixed-effects inference [9].

Methodological Note

Tutor's Methodological Advice: The Carrier Balancing Axiom
The carrier channel in multiplexed single-cell proteomics (SCoPE2) is an analytical double-edged sword. While it provides the critical precursor ion intensity needed to trigger MS2 fragmentation and identify peptide sequences, it directly competes for finite C-trap space-charge capacity. Loading a 500× carrier channel consumes over 98% of instrument charge capacity, compressing single-cell ions to fewer than 5 ions per scan and inflating Poisson measurement error to $> 45\%$. The golden rule of isobaric single-cell design is to calibrate carrier loading strictly between 50× and 100× single-cell equivalents [25, 8, 24].

26.1 Biophysical and Analytical Foundations of Single-Cell Proteomics

26.1.1 The Picogram Challenge: Why Proteomics Cannot Rely on PCR

A single mammalian cell (e.g., a human lymphocyte, fibroblast, or epithelial cell) contains approximately 100 to 200 pg of total protein mass, corresponding to roughly 2×10^9 to 3×10^9 individual polypeptide molecules distributed across 10^4 distinct protein species [28, 1]. In single-cell RNA sequencing (scRNA-seq), picogram amounts of mRNA are reverse-transcribed and amplified through exponential polymerase chain reaction (PCR) cycles to generate microgram quantities of sequencing cDNA [21]. In proteomics, no enzymatic mechanism exists to copy, template, or amplify peptides. Every detected ion represents a physical molecule that survived sample lysis, reduction, alkylation, enzymatic digestion, liquid chromatography, electrospray ionization, ion transmission optics, and mass filtering [19].

26.1.2 Ion Transmission Efficiency and Poisson Shot Noise Limits

Consider a moderately abundant intracellular protein present at $N_{\text{cell}} = 10^5$ copies in a single cell. The number of ions k reaching the mass spectrometer detector is governed by the cumulative transmission efficiency η_{total}:

$$\eta_{\text{total}} = \eta_{\text{prep}} \cdot \eta_{\text{LC}} \cdot \eta_{\text{ESI}} \cdot \eta_{\text{optics}} \cdot \eta_{\text{MS}} \tag{26.1}$$

In state-of-the-art ultra-low-loss microfluidic platforms (e.g., nanoPOTS [18, 10] or acoustic droplet dispensing via cellenONE [33]), total transmission efficiency typically ranges from $\eta_{\text{total}} \approx 0.1\%$ to 1.0% (10^{-3} to 10^{-2}).

Therefore, the expected number of detected ions λ is:

$$\lambda = \mathbb{E}[k] = N_{\text{cell}} \cdot \eta_{\text{total}} = 10^5 \times 0.001 = 100 \text{ ions} \tag{26.2}$$

Ion counting in an orbital ion trap, time-of-flight detector, or electron multiplier follows a Poisson counting process:

$$P(k \mid \lambda) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{26.3}$$

The variance of a Poisson random variable equals its mean ($\text{Var}(k) = \lambda$). Consequently, the theoretical lower bound on technical noise, known as the Poisson shot noise limit, dictates the Coefficient of Variation (CV):

$$\text{CV}_{\text{Poisson}} = \frac{\sqrt{\text{Var}(k)}}{\mathbb{E}[k]} = \frac{\sqrt{\lambda}}{\lambda} = \frac{1}{\sqrt{\lambda}} \tag{26.4}$$

For an abundant protein with $\lambda = 100$ detected ions, the irreducible physical shot noise is $\text{CV} = \frac{1}{\sqrt{100}} = 10\%$. For a low-abundance regulatory protein ($N_{\text{cell}} = 1{,}000$ copies; $\lambda = 1$ ion), $\text{CV} = 100\%$. Technical

measurement variance in single-cell proteomics is physically dictated by the finite quantum nature of ion sampling [26].

26.1.3 Technological Modalities: Isobaric Multiplexing vs. Label-Free DIA

Two primary technological paradigms dominate contemporary single-cell proteomics:

1. **Multiplexed Isobaric Tagging (SCoPE2, plexDIA):** Utilizes tandem mass tags (TMTpro 16-plex or 18-plex) or mTRAQ/plexDIA tags [25, 11]. Multiple single cells are barcoded and pooled into a single LC-MS injection alongside a carrier proteome. This approach maximizes sample throughput (up to 100--200 single cells per day) and provides sequence identification via carrier-boosted precursor signals [22].

2. **Label-Free Single-Cell DIA (timsTOF SCP, Orbitrap Astral):** Advances in high-brightness ion sources, trapped ion mobility spectrometry (TIMS), and ultra-fast mass analyzers (Orbitrap Astral) enable true label-free data-independent acquisition (DIA) from single cells without carrier channels [6, 29, 12, 15]. DIA eliminates carrier ratio compression and isobaric reporter co-isolation interference, achieving quantitative dynamic ranges exceeding 10^4 at the expense of lower sample throughput.

26.2 Multiplexed Isobaric Architectures: The Carrier Proteome Paradigm

To overcome the identification sensitivity barrier in data-dependent acquisition (DDA), Slavov and colleagues developed Single-Cell ProtEomics by Mass Spectrometry (SCoPE-MS [7] and SCoPE2 [25]). SCoPE2 utilizes multiplexed isobaric chemical tags (TMT or TMTpro) to couple single cells to an isobaric ‘‘carrier’’ proteome.

26.2.1 Design Architecture of a Single-Cell TMT Multiplex

A standard 16-plex TMTpro single-cell acquisition incorporates four distinct functional channel classes:

1. **Carrier (Booster) Channel (e.g., TMT-126):** Contains peptide digest equivalent to 50 to 100 cells. In MS1 full scans, the carrier co-elutes with identical single-cell peptides, elevating precursor intensity above the instrument trigger threshold for MS2 fragmentation [25].

2. **Reference Channel (e.g., TMT-127N):** Contains a pooled aliquot (typically 5 cell equivalents) representing all biological conditions in the experiment. The reference channel serves as an invariant internal standard for cross-batch normalization [22].

3. **Single-Cell Channels (e.g., TMT-127C through 134N):** Up to 13 channels dedicated to individual isolated single cells.

4. **Negative Control (Empty/Blank) Channel:** Contains carrier buffer and digestion enzymes without any loaded cell, providing a baseline measurement of non-specific chemical background and isotopic contamination [30].

26.2.2 The Carrier Dilution and Ion Suppression Trade-Off

While the carrier channel enables peptide identification, excessive carrier loading introduces severe quantitative distortion. The C-trap or ion routing multipole of an Orbitrap instrument has a finite total ion capacity

($C_{\text{cap}} \approx 10^5$--$10^6$ charges) to prevent space-charge repulsion and mass shifts:

$$C_{\text{total}} = C_{\text{carrier}} + C_{\text{ref}} + \sum_{j=1}^{m} C_{\text{single},j} \leq C_{\text{cap}} \tag{26.5}$$

If the carrier channel represents a 200-fold excess over individual single cells and the reference channel represents 5 single-cell equivalents:

$$\frac{C_{\text{carrier}}}{C_{\text{total}}} = \frac{C_{\text{carrier}}}{C_{\text{carrier}} + C_{\text{ref}} + \sum_{j=1}^{13} C_{\text{single},j}} = \frac{200}{200+5+13\times 1} = \frac{200}{218} \approx 91.7\% \tag{26.6}$$

Over 90% of the ion trap capacity is consumed exclusively by carrier ions. Single-cell reporter ions are crowded out, reducing the number of sampled single-cell ions λ_{single} and dramatically inflating Poisson shot noise according to Eq. (26.4) [8, 24].

Methodological Pitfall & Deadly Sin

The 200-Cell Carrier Limit:
Studies by Vanderaa, Gatto, and Slavov demonstrated that carrier levels exceeding 200 cell-equivalents saturate the Orbitrap detector dynamic range, exacerbate co-isolation ratio compression, and suppress single-cell reporter ions into electronic baseline noise [30, 8]. The optimal carrier level is strictly 50 to 100 cell-equivalents.

26.3 The scp Data Model: Hierarchical Multi-Assay Architecture

Handling single-cell proteomics requires coordinating multiple levels of biological and chemical granularity: individual PSMs, collapsed peptides, consolidated proteins, across multiple TMT multiplexes and cell annotations [31].

The Bioconductor package `scp` (Vanderaa & Gatto [30, 32]) extends the `QFeatures` container [13] by storing each acquisition multiplex as a dedicated `SingleCellExperiment` (SCE) assay.

The Hierarchical scp Multi-Assay Tree

QFeatures Container

- **Assay 1 (SCE):** `Run01_PSMs` (p_1 PSMs $\times 16$ channels)
- **Assay 2 (SCE):** `Run02_PSMs` (p_2 PSMs $\times 16$ channels)
- ...
- **Joined Assay (SCE):** `all_psms` ($\sum p_i$ PSMs $\times n$ total cells)
- **Aggregated Assay (SCE):** `peptides` (p_{pep} peptides $\times n$ total cells)
- **Protein Assay (SCE):** `proteins` (p_{prot} protein groups $\times n$ total cells)

Figure 26.1: The hierarchical assay topology of `scp`. PSMs from independent TMT runs are linked through feature annotations, aggregated into peptides and proteins, and preserved within a single reproducible container.

The `SingleCellExperiment` assay structure integrates:

- **`assay(sce)`:** Quantitative matrix of reporter ion intensities (rows = features; columns = cells).
- **`colData(sce)`:** Cell-level metadata (batch run, TMT channel, cell type/condition, total intensity, cell diameter).
- **`rowData(sce)`:** Spectral identification metadata (peptide sequence, master protein accession, search engine score, PEP, reverse/contaminant flags).
- **`reducedDims(sce)`:** Dimensionality reduction embeddings (PCA, UMAP, t-SNE) computed on cell profiles.

26.4 Cell Quality Control, Sparsity, and Normalization

26.4.1 Cell-Level Quality Control Metrics

Before biological analysis, single-cell channels must undergo rigorous quality filtering to eliminate technical failures:

1. **Empty Wells / Failed Labeling:** Channels exhibiting median reporter intensity $< 3\times$ the negative control channel or fewer than 300 identified peptides are flagged and discarded [30].
2. **Cellular Doublets:** Channels exhibiting total reporter ion intensity $> 2.5\times$ the median single-cell intensity represent two cells captured within a single well or droplet dispensing error [33].
3. **Median Relative Reporter Ion Intensity (RI):** The ratio of single-cell reporter intensity to the reference channel:
$$\mathrm{RI}_{ij} = \frac{y_{ij}}{y_{i,\mathrm{ref}}} \tag{26.7}$$
High-quality single cells should maintain a stable median relative intensity distribution across the proteome [27].

26.4.2 Leakage-Free Multi-Stage Normalization in SCP

Normalization in multiplexed SCP must correct for two orthogonal technical axes without leaking carrier signal into single-cell estimates:

1. **Intra-Assay Sample Loading Normalization:** Equalizes total protein loading across single-cell channels within a single TMT multiplex. While classic omics workflows employ variance stabilization transformations [16], standard practice in multiplexed single-cell workflows scales channels by total column sums:
$$y_{ij}^{\mathrm{SL}} = \frac{y_{ij}}{\sum_{k=1}^{p} y_{kj}} \times \left(\frac{1}{m} \sum_{c=1}^{m} \sum_{k=1}^{p} y_{kc} \right) \tag{26.8}$$
Alternatively, sample loading can be equalized via median centering ($y_{ij}^{\mathrm{med}} = y_{ij} / \mathrm{median}_k(y_{kj})$), implemented in `QFeatures` via `normalize(..., method = "center.median")` to reduce sensitivity to outlier peptides.

2. **Inter-Batch Internal Reference Scaling (IRS):** Uses the reference channel (present in all multiplexes) to remove run-to-run batch effects across multiple independent TMT plates:

$$\bar{R}_i = \left(\prod_{b=1}^{B} R_{ib} \right)^{1/B} \quad \text{(geometric mean across batches)} \tag{26.9}$$

$$S_{ib} = \frac{\bar{R}_i}{R_{ib}} \tag{26.10}$$

$$y_{ijb}^{\text{IRS}} = y_{ijb}^{\text{SL}} \times S_{ib} \tag{26.11}$$

26.4.3 The Sparsity Fallacy: Why Bulk Imputation Fails in Single-Cell Proteomics

In single-cell proteomics, missing values routinely encompass 40--60% of the data matrix [30, 6]. Missingness is governed by a hybrid biophysical mechanism:

- **Missing Not At Random (MNAR):** Low-abundance peptides whose single-cell reporter ion counts fall below the instrument detection threshold (< 25 ion counts), generating left-censored drop-out.
- **Missing At Random (MAR):** Stochastic precursor selection during DDA duty cycles. Precursor selection probability is governed by precursor ion abundance at the MS1 level; conditioned on the observed precursor abundance (boosted by the carrier channel), selection is stochastic across single-cell channels.

Mathematical Deep Dive

The Danger of Bulk Imputation in SCP:
Applying standard bulk imputation algorithms (e.g., k-nearest neighbors or random forest) across a 50% missing single-cell matrix creates artificial correlations, collapses cell-to-cell variance, and forces cells into spurious synthetic clusters. The gold-standard recommendation in single-cell proteomics is to perform dimensionality reduction and differential abundance directly on filtered, non-imputed matrices, or utilize model-based approaches (such as Hurdle models or mixed-effects models) that account for missingness explicitly [9, 2, 3].

26.5 Hands-On R Laboratory: The Complete scp Workflow

We now construct an end-to-end, reproducible R laboratory using the `scp` and `QFeatures` architecture. We simulate a multi-channel single-cell proteomics experiment (13 single cells across two biological states, 1 carrier channel, 1 reference channel, and 1 blank channel), audit cell QC, filter contaminants, normalize, aggregate features, and perform PCA visualization.

R Lab Listing 1: Simulating a 16-Plex Single-Cell TMT Multiplex [R 4.4 / Bioconductor]

```
# ------------------------------------------------------------------------
# Simulation of a 16-Plex TMTpro Single-Cell Experiment
# Channels: 1 Carrier (100x), 1 Reference (5x), 1 Empty/Blank, 13 Single Cells
# Single Cells: 7 Naive Stem Cells, 6 Primed Stem Cells
# ------------------------------------------------------------------------
set.seed(303)

n_psms <- 800
```

```
channels <- c("TMT_126_Carrier", "TMT_127N_Ref", "TMT_127C_Blank",
              paste0("TMT_Naive_", 1:7), paste0("TMT_Primed_", 1:6))
n_channels <- length(channels)

# Base log2 peptide abundance across proteome (mean = 11.0, sd = 1.8)
base_abund <- rnorm(n_psms, mean = 11.0, sd = 1.8)

# True biology: 80 PSMs (first 16 proteins) up-regulated in Naive cells (FC = 2.5, log2FC = +1.32)
bio_effect <- rep(0, n_psms)
bio_effect[1:80] <- 1.32

# Channel scaling factors (relative loading)
channel_loadings <- c(100, 5, 0.0, rep(1.0, 7), rep(1.0, 6))

raw_psm_mat <- matrix(NA_real_, nrow = n_psms, ncol = n_channels)
colnames(raw_psm_mat) <- channels
rownames(raw_psm_mat) <- paste0("PSM_", sprintf("%04d", 1:n_psms))

# Carrier (100x) and Reference (5x) channels
raw_psm_mat[, 1] <- rpois(n_psms, lambda = (2^base_abund) * 100)
raw_psm_mat[, 2] <- rpois(n_psms, lambda = (2^base_abund) * 5)

# Negative control / blank channel: low Poisson background noise
blank_counts <- rpois(n_psms, lambda = 2)
raw_psm_mat[, 3] <- ifelse(blank_counts < 8, NA, blank_counts)

# Single cells: 1x loading with physical Poisson shot noise and detection limits
lod_thresh <- 25  # Limit of Detection in raw ion counts

for (j in 4:n_channels) {
  is_naive <- grepl("Naive", channels[j])
  exp_log2 <- base_abund + (if (is_naive) bio_effect else 0)
  exp_linear <- 2^exp_log2

  # Physical Poisson ion arrival counts
  counts <- rpois(n_psms, lambda = exp_linear)

  # Intensity-dependent DDA precursor sampling (MAR)
  prob_detect <- 1 / (1 + exp(-0.8 * (exp_log2 - 10.5)))
  selected <- runif(n_psms) < prob_detect

  # Apply limit of detection (MNAR) and duty cycle sampling
  obs <- counts
  obs[!selected | counts < lod_thresh] <- NA
  raw_psm_mat[, j] <- obs
}

# Construct Channel Metadata (colData)
col_meta <- data.frame(
  Channel   = channels,
  CellType  = c("Carrier", "Reference", "Blank", rep("Naive", 7), rep("Primed", 6)),
  WellIndex = 1:16,
  row.names = channels,
  stringsAsFactors = FALSE
)

# Construct Strictly Nested Feature Metadata (rowData): 160 proteins -> 320 peptides -> 800 PSMs
```

```
pep_ids <- paste0("PEP_", sprintf("%03d", rep(1:320, times = rep(c(3, 2), times = 160))))
prot_ids <- rep(paste0("PROT_", sprintf("%03d", 1:160)), each = 5)

row_meta <- data.frame(
  PSM_ID   = rownames(raw_psm_mat),
  Protein  = prot_ids,
  Peptide  = pep_ids,
  PEP      = runif(n_psms, 0.001, 0.08),
  Reverse  = sample(c(rep("-", 760), rep("+", 40))),
  stringsAsFactors = FALSE
)

cat(sprintf("Simulated %d PSMs mapping to 320 peptides and 160 proteins.\n", n_psms))
cat(sprintf("Carrier Intensity: %.2e | Median Single-Cell Intensity: %.2e\n",
            median(raw_psm_mat[, 1], na.rm = TRUE),
            median(raw_psm_mat[, 4:16], na.rm = TRUE)))
```

```
Simulated 800 PSMs mapping to 320 peptides and 160 proteins.  Carrier Intensity:  2.12e+05 | Median Single-Cell
Intensity: 3.82e+03
```

R Lab Listing 2: Building the QFeatures Single-Cell Container [R 4.4 / Bioconductor]

```
# ------------------------------------------------------------------------------
# Construct SingleCellExperiment and QFeatures Hierarchy
# ------------------------------------------------------------------------------
library(SingleCellExperiment)
library(QFeatures)

sce_psm <- SingleCellExperiment(
  assays  = list(psms_raw = raw_psm_mat),
  rowData = row_meta,
  colData = col_meta
)

qf <- QFeatures(list(psms = sce_psm), colData = col_meta)

cat("QFeatures Container Structure:\n")
print(qf)
```

```
QFeatures Container Structure: An instance of class QFeatures (type: scp) with 1 set:
[1] psms: SingleCellExperiment with 800 rows and 16 columns
```

R Lab Listing 3: Cell-Level Quality Control and Filtering [R 4.4 / Bioconductor]

```
# ------------------------------------------------------------------------------
# Quality Control: Filter Contaminants and Exclude Carrier/Blank Channels
# ------------------------------------------------------------------------------
# 1. Feature Filtering: Remove Decoys (Reverse == '+') and high PEP (> 0.05)
qf <- filterFeatures(qf, ~ Reverse == "-" & PEP < 0.05)

cat(sprintf("PSMs retained after identification filtering: %d / 800\n",
            nrow(qf[["psms"]])))
```

```
# 2. Cell Filtering: Retain Single-Cell Channels for Biological Quantification
qf_single <- subsetByColData(qf, colData(qf)$CellType %in% c("Naive", "Primed"))

cat(sprintf("Channels retained for single-cell quantification: %d\n",
            ncol(qf_single[["psms"]])))
```

```
PSMs retained after identification filtering: 475 / 800 Channels retained for single-cell quantification: 13
```

Note on Reference Channels: In this single-batch demonstration, the reference channel is excluded alongside carrier and blank channels prior to single-cell quantification because cross-batch IRS normalization requires multiple acquisition batches. In multi-batch studies, the reference channel is preserved across all assays to calculate inter-batch scaling factors before subsetting single cells.

R Lab Listing 4: Multi-Level Aggregation and Normalization [R 4.4 / Bioconductor]

```
# ------------------------------------------------------------------------------
# Feature Aggregation: PSM -> Peptide -> Protein
# Normalization: Median Centering across Single Cells
# ------------------------------------------------------------------------------
# Step 1: Remove unobserved features across single cells before robust aggregation
psm_obs <- rowSums(!is.na(assay(qf_single[["psms"]])))
qf_single <- qf_single[names(which(psm_obs >= 1)), , ]

# Step 2: Normalize PSM intensities across single cells (median centering)
qf_single <- normalize(qf_single,
                       i = "psms",
                       name = "psms_norm",
                       method = "center.median")

# Step 3: Aggregate PSMs to Peptides (Robust Tukey Biweight M-estimation)
qf_single <- aggregateFeatures(qf_single,
                               i = "psms_norm",
                               fcol = "Peptide",
                               name = "peptides",
                               fun = MsCoreUtils::robustSummary)

# Step 4: Aggregate Peptides to Protein Groups
qf_single <- aggregateFeatures(qf_single,
                               i = "peptides",
                               fcol = "Protein",
                               name = "proteins",
                               fun = MsCoreUtils::robustSummary)

cat("\nAggregated QFeatures Assay Topology:\n")
print(names(qf_single))
cat(sprintf("Retained Features: %d peptides | %d proteins\n",
            nrow(qf_single[["peptides"]]), nrow(qf_single[["proteins"]])))
cat(sprintf("Final Protein Matrix Dimensions: %d proteins x %d single cells\n",
            nrow(qf_single[["proteins"]]), ncol(qf_single[["proteins"]])))
```

```
Aggregated QFeatures Assay Topology: [1] "psms" "psms_norm" "peptides" "proteins" Retained Features: 286 peptides
| 158 proteins Final Protein Matrix Dimensions: 158 proteins x 13 single cells
```

R Lab Listing 5: Single-Cell Phenotypic Clustering via Principal Component Analysis [R 4.4 / Bioconductor]

```r
# ------------------------------------------------------------------------------
# PCA on Normalized Protein Matrix to Evaluate Cell State Recovery
# ------------------------------------------------------------------------------
prot_matrix <- assay(qf_single[["proteins"]])

# Restrict PCA to complete-case proteins (standard non-imputed single-cell analysis)
prot_complete <- na.omit(prot_matrix)
cat(sprintf("Complete-case proteins for PCA: %d / %d\n",
            nrow(prot_complete), nrow(prot_matrix)))

# Perform PCA on transposed matrix (cells as rows)
pca_res <- prcomp(t(prot_complete), scale. = TRUE)
pca_var <- round(summary(pca_res)$importance[2, 1:2] * 100, 1)

# Annotate with Cell Type
cell_types <- colData(qf_single[["proteins"]])$CellType
pc1_scores <- pca_res$x[, 1]

# Assess linear separation between Naive and Primed
t_test_res <- t.test(pc1_scores ~ cell_types)

cat("--- Single-Cell Phenotype Recovery Benchmark ---\n")
cat(sprintf("PC1 Variance Explained: %.1f%% | PC2 Variance: %.1f%%\n",
            pca_var[1], pca_var[2]))
cat(sprintf("PC1 Separation between Naive and Primed Cells: t = %.2f (p = %.2e)\n",
            t_test_res$statistic, t_test_res$p.value))
```

```
Complete-case proteins for PCA: 89 / 158 — Single-Cell Phenotype Recovery Benchmark — PC1 Variance Explained: 85.6%
| PC2 Variance: 9.7% PC1 Separation between Naive and Primed Cells: t = 2.41 (p = 4.12e-02)
```

Biological Discovery & Validation

Interpretation of Single-Cell Separation:
Despite 40% identification sparsity and Poisson shot noise, Tukey biweight M-estimation and non-imputed complete-case PCA cleanly recover biological cell states along PC1 ($t = 2.41, p = 0.041$), explaining 85.6% of total proteomic variance across complete cases.

26.6 The Five Deadly Sins of Single-Cell Proteomics

Methodological Pitfall & Deadly Sin

Deadly Sin 1: Excessive Carrier Channel Loading ($> 200\times$ Booster)
Mechanism: Spiking in a 500-cell or 1,000-cell equivalent carrier channel to maximize the total number of peptide identifications [25, 8].
Mathematical Consequence: Ion trap space-charge saturation. Over 98% of trapped ions belong to the

carrier, compressing single-cell reporter ion counts down to 0--10 ions per scan. Quantitative precision collapses due to severe Poisson shot noise (CV $> 40\%$).
Remedy: Restrict carrier channels to $50\times$--$100\times$ single-cell equivalents [30].

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Imputing High-Sparsity SCP Matrices with Bulk MCAR Algorithms
Mechanism: Applying k-nearest neighbors (KNN) or Random Forest imputation across an SCP matrix with $> 50\%$ missing values [6].
Mathematical Consequence: Bulk imputation assumes low-percentage missingness. In ultra-sparse matrices, KNN connects cells based on missingness patterns rather than authentic abundance, generating artificial clusters that mimic novel cell sub-types.
Remedy: Filter features by minimum detection threshold ($> 50\%$ across cells) and utilize model-based statistical frameworks (such as mixed-effects models) that handle missingness naturally [9].

Methodological Pitfall & Deadly Sin

Deadly Sin 3: The Pseudoreplication Fallacy (Cells as Biological Replicates)
Mechanism: Treating 50 single cells isolated from a single cell culture dish or a single patient biopsy as 50 independent biological replicates in a two-sample t-test [2, 3].
Mathematical Consequence: Severe Type I error inflation (p-values under-estimated by orders of magnitude). Cells within the same culture dish share identical micro-environmental, media, and passage history.
Remedy: Employ hierarchical linear mixed models ($\sim \text{Condition} + (1 \mid \text{BiologicalReplicate}) + (1 \mid \text{Run})$) to account for correlation among cells from the same origin [4].

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Normalizing Single Cells Without Excluding the Carrier Channel
Mechanism: Applying global column sum or median normalization to a raw 16-plex TMT assay that still includes the carrier channel [32].
Mathematical Consequence: Because the carrier contains 100-fold more signal than single cells, global normalization algorithms align single-cell intensities to carrier fluctuations, distorting single-cell abundance measurements.
Remedy: Exclude carrier and empty channels using `subsetByColData()` prior to calculating single-cell normalization factors [22].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Clustering Unnormalized Raw Reporter Intensities
Mechanism: Executing PCA, UMAP, or Leiden clustering directly on raw single-cell reporter ion intensities without sample loading adjustment [27].
Mathematical Consequence: PC1 reflects total cell volume, pipetting volume, or channel labeling efficiency rather than biological cell identity. Cells cluster by acquisition run or cell diameter.
Remedy: Normalize by total cell reporter sum or median, log2-transform, and center features prior to

dimensionality reduction [31].

26.7 Tutor's Self-Assessment Diagnostic Checklist

Table 26.2: Tutor's Self-Assessment Diagnostic Checklist for Single-Cell Proteomics Workflows

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Carrier Ratio Balance	Carrier load between 50×--100× single-cell equivalents [25].	If $>200\times$: Reduce carrier digest load to eliminate ion suppression and ratio compression.
2	Single-Cell Signal/Blank	Median single-cell reporter intensity $\geq 3\times$ blank/empty channel [30].	If <3.0: Flag channel as failed cell capture or incomplete enzymatic lysis; discard channel.
3	Identification Coverage	≥ 300 peptides / ≥ 150 protein groups per single cell [23].	If below threshold: Inspect acoustic droplet dispensing efficiency or microfluidic chip clog.
4	Doublet Detection Ratio	Channel total signal $\leq 2.0\times$ median single-cell signal [33].	If $>2.5\times$: Flag as cellular doublet (two cells captured in single droplet); remove channel.
5	Missing Data Integrity	Matrix sparsity handled via filtering ($>50\%$ non-NA) without global KNN imputation [6].	If imputed with bulk algorithms: Retain raw missing patterns; fit hurdle or mixed models [9].
6	Replicate Independence	Multi-cell experiments modeled via hierarchical mixed models [2, 3].	If analyzed via naive t-test: Add donor/batch random intercepts to control Type I error.

26.8 Chapter Exercises

Exercise 26.1 (Theoretical Limits of Poisson Ion Sampling):
A low-abundance signaling kinase is expressed at $N_{\text{cell}} = 2{,}500$ copies in a single mammalian cell. An analytical SCP workflow achieves an overall ion transmission efficiency of $\eta_{\text{total}} = 0.40\%$.

1. Calculate the expected number of ions λ reaching the mass spectrometer detector.

2. What is the theoretical minimum Coefficient of Variation ($\text{CV}_{\text{Poisson}}$) achievable for this protein?

3. If carrier ion influx reaches the automatic gain control (AGC) target rapidly, reducing single-cell ion accumulation time to 25% of the single-cell-only fill time ($t'_{\text{fill}} = 0.25\,t_{\text{fill}}$) to prevent space-charge saturation, calculate the new expected ion count λ' and revised CV'.

Exercise 26.2 (The SCoPE2 Carrier Bias Equation):
Let I_{carrier} be the reporter intensity of the carrier channel, and let I_1 and I_2 be the true reporter intensities of

two single cells with true fold change $R_{\text{true}} = I_1/I_2 = 3.0$. Suppose co-isolation of precursor ions introduces 15% carrier contamination into both single-cell reporter windows:

$$I_1^{\text{obs}} = I_1 + 0.15\, I_{\text{carrier}}, \quad I_2^{\text{obs}} = I_2 + 0.15\, I_{\text{carrier}} \tag{26.12}$$

If the carrier is loaded at 100-fold excess relative to cell 2 ($I_{\text{carrier}} = 100 I_2$), calculate the observed fold change $R_{\text{obs}} = I_1^{\text{obs}}/I_2^{\text{obs}}$. What is the percentage ratio compression?

Exercise 26.3 (Single-Cell Quality Control Filter Function in R):
Write an R function `audit_scp_channels(assay_mat, blank_col, min_psms, max_ratio)` that computes per-channel total intensity, identified feature count, and ratio over the blank channel. Flag channels with channel-type awareness as ‘‘CONTROL_BLANK’’, ‘‘CONTROL_CARRIER’’, ‘‘CONTROL_REFERENCE’’, ‘‘FLAG_EMPTY_WELL’’, ‘‘FLAG_DOUBLET’’, or ‘‘PASS’’.

Exercise 26.4 (Hierarchical Mixed Model vs. Naive t-Test on Single Cells):
Write an R simulation comparing a naive Student’s t-test against a donor-level paired analysis on 60 single cells derived from 6 biological donor replicates across two conditions (5 cells in Condition A and 5 cells in Condition B per donor). Demonstrate that when intra-donor and culture preparation correlation is present (ICC $= 0.35$), the naive t-test produces an inflated false positive rate ($> 25\%$) on null data, while the donor-level analysis preserves the nominal $\alpha = 0.05$ error rate.

26.9 Worked-Out Solutions

26.9.1 Solution to Exercise 26.1: Poisson Sampling Limits

1. Expected Number of Detected Ions:
Given $N_{\text{cell}} = 2{,}500$ copies and $\eta_{\text{total}} = 0.40\% = 0.0040$:

$$\lambda = N_{\text{cell}} \cdot \eta_{\text{total}} = 2{,}500 \times 0.0040 = 10 \text{ ions} \tag{26.13}$$

2. Theoretical Minimum Poisson CV:
From Eq. (26.4):

$$\text{CV}_{\text{Poisson}} = \frac{1}{\sqrt{\lambda}} = \frac{1}{\sqrt{10}} \approx 0.3162\ (31.62\%) \tag{26.14}$$

Even under perfectly noiseless electronics, the physical quantum measurement uncertainty is 31.6%.

3. Impact of Carrier Channel Fill-Time Truncation:
When carrier ion influx causes the instrument automatic gain control (AGC) target to be reached rapidly, the single-cell ion accumulation time is truncated to $t'_{\text{fill}} = 0.25\, t_{\text{fill}}$ to prevent space-charge repulsion. The expected single-cell ion count scales linearly with fill time:

$$\lambda' = 0.25 \times \lambda = 0.25 \times 10 = 2.5 \text{ ions} \tag{26.15}$$

The revised theoretical Poisson CV is:

$$\text{CV}' = \frac{1}{\sqrt{\lambda'}} = \frac{1}{\sqrt{2.5}} \approx 0.6325\ (63.25\%) \tag{26.16}$$

Carrier ion crowding reduced single-cell ion sampling by 4-fold, inflating measurement uncertainty from 31.6% to 63.3%. ■

26.9.2 Solution to Exercise 26.2: SCoPE2 Carrier Ratio Compression

Given: $R_{\text{true}} = I_1/I_2 = 3.0 \implies I_1 = 3.0I_2$. Carrier loading: $I_{\text{carrier}} = 100I_2$. Contamination fraction: $f = 0.15$.

1. Observed Reporter Intensities:

$$I_1^{\text{obs}} = I_1 + 0.15\,I_{\text{carrier}} = 3.0I_2 + 0.15(100I_2) = 3.0I_2 + 15.0I_2 = 18.0I_2 \tag{26.17}$$

$$I_2^{\text{obs}} = I_2 + 0.15\,I_{\text{carrier}} = 1.0I_2 + 0.15(100I_2) = 1.0I_2 + 15.0I_2 = 16.0I_2 \tag{26.18}$$

2. Observed Ratio:

$$R_{\text{obs}} = \frac{I_1^{\text{obs}}}{I_2^{\text{obs}}} = \frac{18.0I_2}{16.0I_2} = \frac{18.0}{16.0} = 1.125 \tag{26.19}$$

The true fold change of 3.00 ($\log_2 \text{FC} = 1.585$) was compressed to 1.125 ($\log_2 \text{FC} = 0.170$).

3. Percentage Signal Compression:

$$\text{Compression} = \frac{R_{\text{true}} - R_{\text{obs}}}{R_{\text{true}} - 1} \times 100\% = \frac{3.00 - 1.125}{3.00 - 1} \times 100\% = \frac{1.875}{2.00} \times 100\% = 93.75\% \tag{26.20}$$

Over 93.7% of the authentic biological fold change was obliterated by carrier co-isolation. ■

26.9.3 Solution to Exercise 26.3: Single-Cell QC Audit Function in R

R Lab Listing 6: Automated Single-Cell QC Channel Auditing [R 4.4 / Bioconductor]

```r
audit_scp_channels <- function(assay_mat, blank_col = "TMT_127C_Blank",
                               min_psms = 250, max_ratio = 2.5) {
  # 1. Blank Channel Intensity Baseline
  blank_intensity <- median(assay_mat[, blank_col], na.rm = TRUE)
  if (is.na(blank_intensity) || blank_intensity == 0) blank_intensity <- 1.0

  # 2. Per-Channel Metrics
  tot_intensity <- colSums(assay_mat, na.rm = TRUE)
  med_intensity <- apply(assay_mat, 2, median, na.rm = TRUE)
  n_detected    <- colSums(!is.na(assay_mat) & assay_mat > 0)

  # Identify Channel Classes
  is_blank   <- colnames(assay_mat) == blank_col
  is_carrier <- grepl("Carrier", colnames(assay_mat), ignore.case = TRUE)
  is_ref     <- grepl("Ref", colnames(assay_mat), ignore.case = TRUE)
  is_single  <- !is_blank & !is_carrier & !is_ref

  # Median single-cell intensity across bona fide single cells
  median_sc_tot <- median(tot_intensity[is_single], na.rm = TRUE)

  signal_to_blank <- med_intensity / blank_intensity
  doublet_ratio   <- tot_intensity / median_sc_tot

  # 3. Classify Channel Status with Channel-Type Awareness
  status <- rep("PASS", ncol(assay_mat))
  status[is_blank]   <- "CONTROL_BLANK"
  status[is_carrier] <- "CONTROL_CARRIER"
  status[is_ref]     <- "CONTROL_REFERENCE"
```

```
  # Single-cell quality control filters
  status[is_single & (n_detected < min_psms | signal_to_blank < 3.0)] <- "FLAG_EMPTY_WELL"
  status[is_single & doublet_ratio > max_ratio] <- "FLAG_DOUBLET"

  qc_summary <- data.frame(
    Channel         = colnames(assay_mat),
    Features_Count  = n_detected,
    Total_Signal    = tot_intensity,
    Signal_To_Blank = round(signal_to_blank, 2),
    Doublet_Ratio   = round(doublet_ratio, 2),
    Status          = status,
    row.names       = NULL,
    stringsAsFactors = FALSE
  )

  cat("--- Single-Cell Channel Quality Control Summary ---\n")
  print(qc_summary)
  return(invisible(qc_summary))
}

# Test on simulated PSM matrix from Section 26.5
qc_res <- audit_scp_channels(raw_psm_mat, blank_col = "TMT_127C_Blank")
```

```
— Single-Cell Channel Quality Control Summary — Channel Features_Count Total_Signal Signal_To_Blank Doublet_Ratio
Status 1 TMT_126_Carrier 800 358685739 26439.19 104.19 CONTROL_CARRIER 2 TMT_127N_Ref 800 17936194 1334.62 5.21
CONTROL_REFERENCE 3 TMT_127C_Blank 2 16 1.00 0.00 CONTROL_BLANK 4 TMT_Naive_1 500 3587542 470.44 1.04 PASS 5
TMT_Naive_2 479 3442603 498.12 1.00 PASS 6 TMT_Naive_3 489 3569712 480.50 1.04 PASS 7 TMT_Naive_4 468 3521952
502.25 1.02 PASS 8 TMT_Naive_5 491 3573402 486.75 1.04 PASS 9 TMT_Naive_6 482 3513612 500.94 1.02 PASS 10 TMT_Naive_7
483 3589663 493.88 1.04 PASS 11 TMT_Primed_1 464 3028614 488.62 0.88 PASS 12 TMT_Primed_2 473 3028219 448.00 0.88
PASS 13 TMT_Primed_3 471 2987459 454.62 0.87 PASS 14 TMT_Primed_4 461 2935309 463.88 0.85 PASS 15 TMT_Primed_5 472
2952121 413.75 0.86 PASS 16 TMT_Primed_6 472 2973822 477.56 0.86 PASS
```

26.9.4 Solution to Exercise 26.4: Hierarchical Mixed Model vs. Naive t-Test

R Lab Listing 7: Demonstration of Pseudoreplication and Type I Error Inflation [R 4.4 / Bioconductor]

```
# Crossed Experimental Design: 6 Donors, each tested in both Condition A and Condition B
# 10 Single Cells per donor (5 cells Condition A, 5 cells Condition B) -> 60 total cells
set.seed(999)

n_donors <- 6
cells_per_cond <- 5
cells_per_donor <- cells_per_cond * 2
n_total_cells <- n_donors * cells_per_donor

donor_ids <- rep(paste0("Donor_", 1:n_donors), each = cells_per_donor)
conditions <- rep(rep(c("Condition_A", "Condition_B"), each = cells_per_cond), times = n_donors)
# Distinct culture preparations / wells per donor-condition pair
well_ids <- rep(paste0("Well_", 1:(n_donors * 2)), each = cells_per_cond)

# 200 Null Simulations
p_naive_list <- numeric(200)
p_lmm_list   <- numeric(200)
```

```
for (sim in 1:200) {
  # Biological donor baseline variance (ICC_donor = 0.35)
  u_donor <- rep(rnorm(n_donors, mean = 0, sd = 1.0), each = cells_per_donor)
  # Culture well preparation variance (ICC_well = 0.30)
  u_well  <- rep(rnorm(n_donors * 2, mean = 0, sd = 0.9), each = cells_per_cond)
  cell_noise <- rnorm(n_total_cells, mean = 0, sd = 0.7)

  y_null <- 10 + u_donor + u_well + cell_noise

  # 1. Naive two-sample t-test (pseudoreplication: treating 60 cells as independent)
  p_naive <- t.test(y_null ~ conditions)$p.value
  p_naive_list[sim] <- p_naive

  # 2. Donor-level paired analysis (pseudo-bulk aggregation per well)
  df_sim <- data.frame(y = y_null, condition = conditions, donor = donor_ids)
  df_agg <- aggregate(y ~ condition + donor, data = df_sim, FUN = mean)
  y_a <- df_agg$y[df_agg$condition == "Condition_A"]
  y_b <- df_agg$y[df_agg$condition == "Condition_B"]
  p_lmm <- t.test(y_a, y_b, paired = TRUE)$p.value
  p_lmm_list[sim] <- p_lmm
}

false_pos_naive <- mean(p_naive_list < 0.05) * 100
false_pos_lmm   <- mean(p_lmm_list < 0.05) * 100

cat("--- Pseudoreplication Type I Error Audit (200 Null Simulations) ---\n")
cat(sprintf("Naive Two-Sample t-Test False Positive Rate: %.1f%% (Catastrophic Error Inflation!)\n",
            false_pos_naive))
cat(sprintf("Donor-Level Paired Analysis False Positive Rate: %.1f%% (Preserves alpha = 5%%)\n",
            false_pos_lmm))
```

```
— Pseudoreplication Type I Error Audit (200 Null Simulations) — Naive Two-Sample t-Test False Positive Rate: 25.5%
(Catastrophic Error Inflation!) Donor-Level Paired Analysis False Positive Rate: 4.0% (Preserves alpha = 5%)
```

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